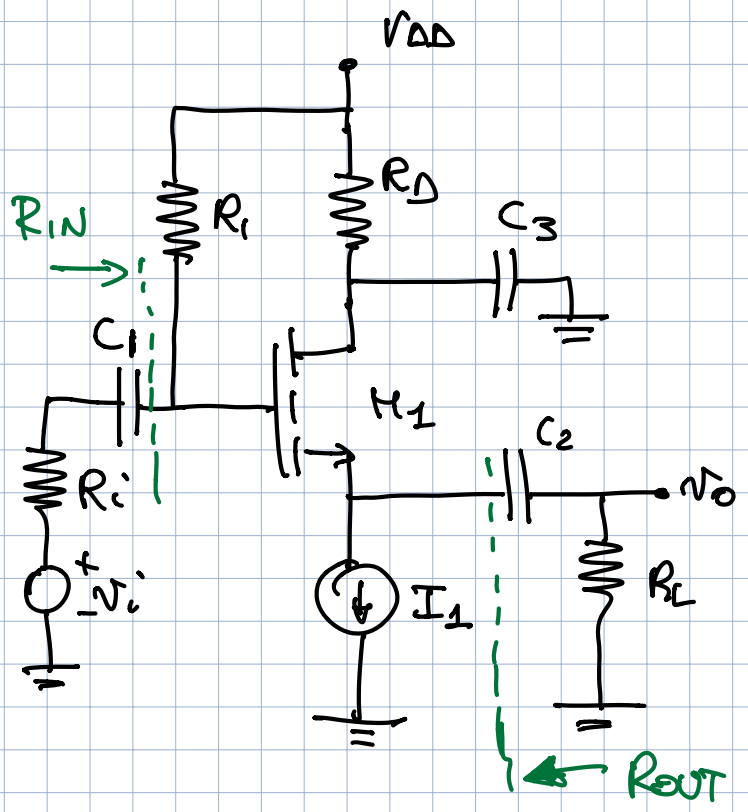
ESERCIZIO MOSFET A3



$V_{DD} = 12V$

$I_1 = 125\ mA$

$R_1 = 1\ M\Omega$

$R_D = 16\ \Omega$ (N.B.)

$R_i = 100\ K\Omega$

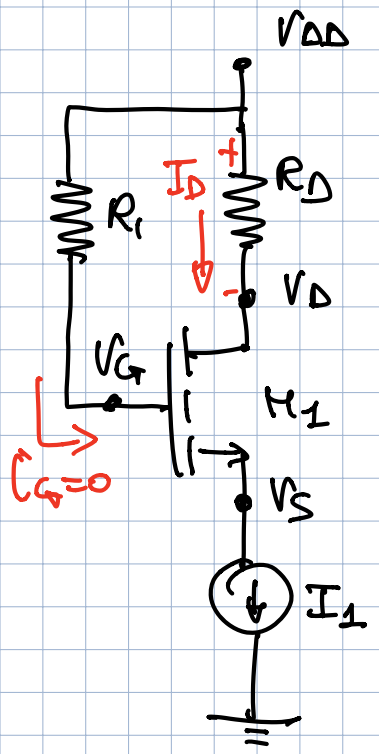
$R_L = 100\ \Omega$

M_1: $V_{TN} = 2V$

$K_n = 10\ mA/V^2$

$\lambda = 10^{-3}\ V^{-1}$

1) ANALISI CN DC



$I_D = I_1 = 125\ mA$

$$I_D = \frac{K_n}{2}(V_{GS} - V_{TN})^2$$

$$V_{GS} = V_{TN} \pm \sqrt{\frac{2I_D}{K_n}} = 2V \pm 5V$$

$V_{GS} \to 7V$ OK $> V_{TN}$

$V_{GS} \to -3V$ NO!

$i_G = 0 \Rightarrow V_G = V_{DD} = 12V$

$V_S = V_G - V_{GS} = 12V - 7V = 5V$

$V_{DS} = V_{DD} - R_D I_D - V_S$

$= 12V - 2V - 5V = 5V$ $\qquad V_{DS} > V_{GS} - V_{TN} = 5V$

MOSFET TRA TRIODO e SATUR.

PONGO $R_D = 0 \Rightarrow V_D = V_{DD}$ $\quad$ $V_{DS} = V_{DD} - V_S = 7V$

$M_1 \left[I_D = 125\ mA,\ V_{DS} = 7V \right]$

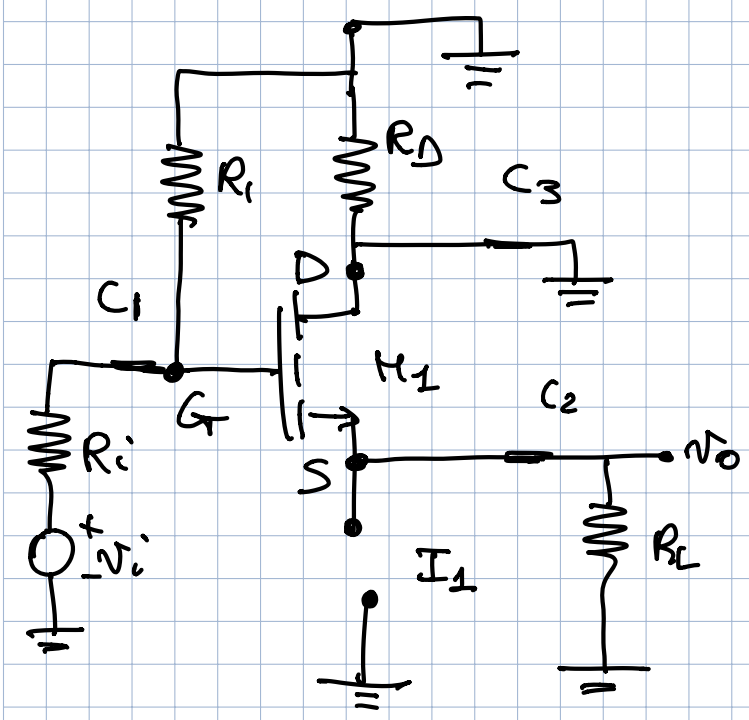
2) PARAMETRI AL PICCOLO SEGNALE

$$g_m = \frac{2 I_D}{V_{GS} - V_{TN}} = \frac{250\ mA}{5V} = 50\ mS$$

$$r_o = \frac{\frac{1}{\lambda} + V_{DS}}{I_D} = \frac{(1000 + 7)V}{125\ mA} = 8{,}056\ K\Omega$$

$r_o \simeq 8\ K\Omega$

3) ANALISI AL PICCOLO SEGNALE

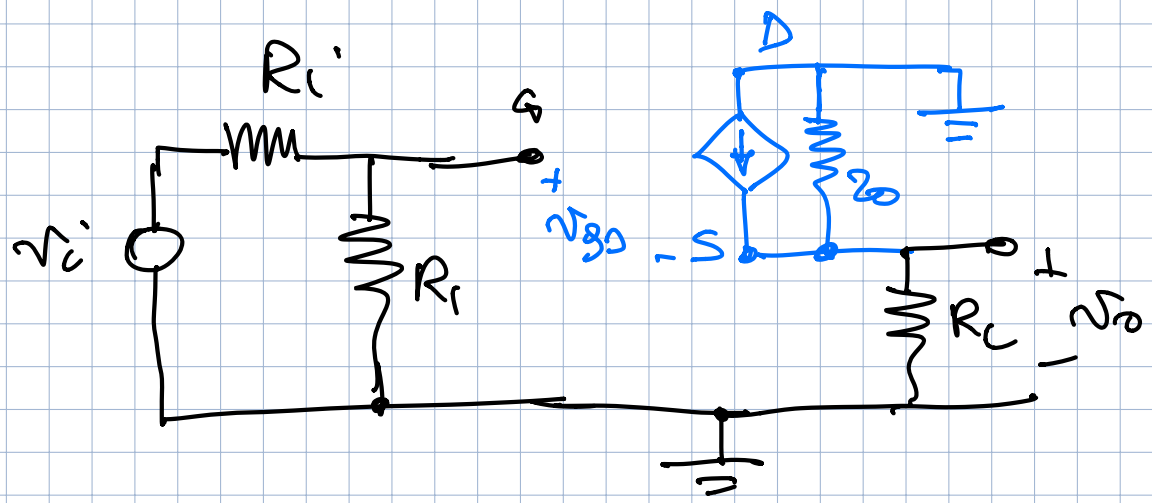


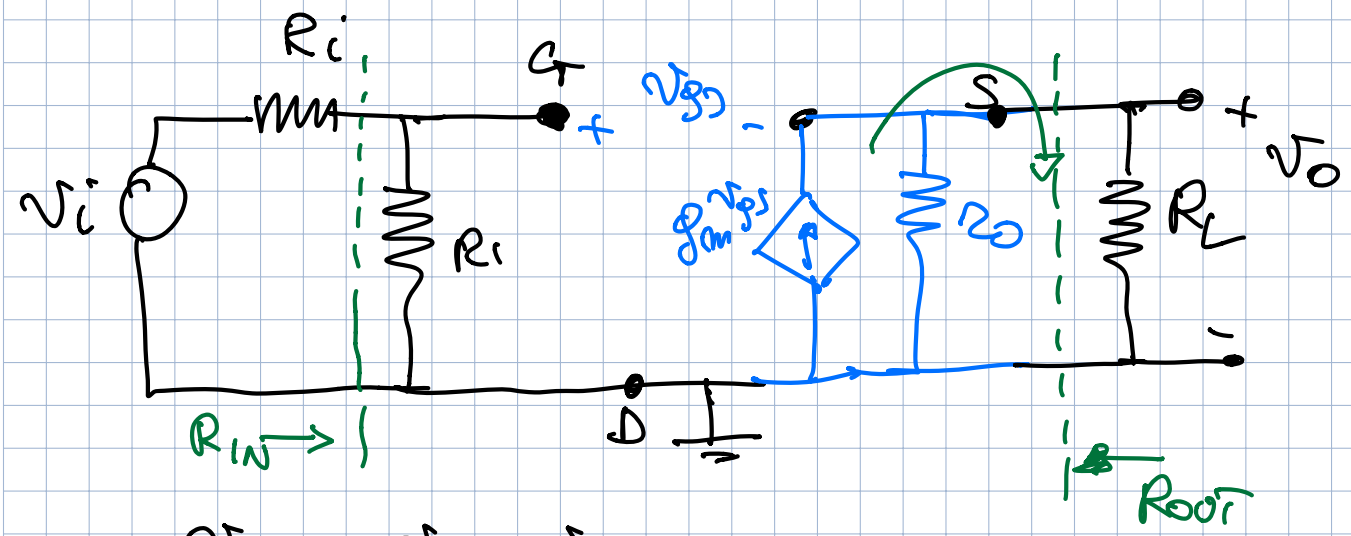
N.B.

GEN. DI CORRENTE APERTO

GEN. TENSIONE $V = 0$

CONDENSATORI C.C. (INDUTTORI C.A).





$$A_v = \frac{v_o}{v_i} = \frac{v_o}{v_G} \cdot \frac{v_G}{v_i}$$

$$v_o = g_m v_{gs}\, r_o // R_L$$

$$v_G = v_{gs} + g_m v_{gs}\, r_o // R_L$$

$$= v_{gs}\left[1 + g_m r_o // R_L\right]$$

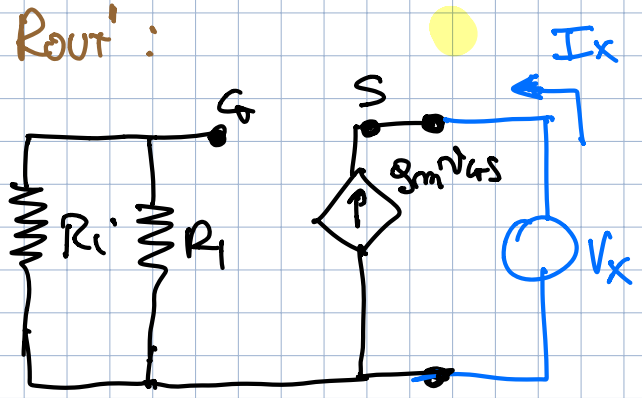
$$\Rightarrow v_{gs} = \frac{v_G}{1 + g_m r_o // R_L}$$

$$\frac{v_o}{v_G} = \frac{g_m r_o // R_L}{1 + g_m r_o // R_L} = 0{,}832$$

$$\frac{v_G}{v_i} = \frac{R_1}{R_1 + R_i} = 0{,}91 \qquad \Rightarrow A_v = \frac{v_o}{v_i} = 0{,}757$$

$R_{IN} = R_1 = 1\ M\Omega$ $\qquad$ R_{OUT} ? $\qquad$ $R_{OUT} = r_o // R_{OUT}'$

R_{OUT}' :



$V_G = 0 \Rightarrow I_x = -g_m v_{GS}$

$v_{GS} = v_G - v_S = -v_S$

$v_S = v_x$

$\Rightarrow I_x = g_m v_x$

$$\Rightarrow R_{OUT}' = \frac{v_x}{I_x} = \frac{1}{g_m} = 20\ \Omega$$

$$R_{OUT} = r_o // R_{OUT}' = 19{,}95\ \Omega$$